Stage 5: Clinical Information Extraction with BioBERT

Theory notes, model choice and interview questions
Project: Doctor–Patient Conversation Audio -> SOAP Notes

What this stage does: runs BioBERT token-classification (NER) on every DOCTOR / PATIENT utterance and extracts clinical entities with exact character positions, confidence and the speaker who said them.

Two versions were built:
Version 1 - public fine-tuned BioBERT checkpoints (BC5CDR): labels DISEASE and CHEMICAL only; no duration or severity.
Version 2 (default) - **we fine-tuned base BioBERT ourselves** on MACCROBAT2018: labels Sign_symptom, Disease_disorder, Medication, Duration, Severity, Dosage, Frequency. Test F1 0.58.

Contents

1. What Named Entity Recognition is
2. BERT in one page
3. What makes BioBERT different
4. From BioBERT to an NER model: fine-tuning
5. BIO tagging and sub-word tokens
6. Our pipeline, step by step
7. Model selection: what exists on Hugging Face
8. Version 1 label mismatch: what we can and cannot extract
9. Version 1 results
10. Evaluation of NER
11. Limitations
12. Version 2: fine-tuning BioBERT ourselves on MACCROBAT2018 (Approach B)
13. Alternatives to BioBERT
14. Interview questions and answers

1. What Named Entity Recognition (NER) is

NER finds spans of text that refer to things of a given type and labels them:

```
I have had  fever  for three days and a severe  headache .
          [DISEASE]                            [DISEASE]
```

- It is a **token classification** task: every token receives a label; consecutive labelled tokens form an entity.
- Clinical NER is the first step of turning free conversation into structured data (symptoms, drugs, conditions) that a SOAP note can be built from.
- NER only **finds** mentions. It does not decide if a symptom is present, absent ('no fever') or belongs to someone else - those are separate tasks (negation / assertion detection).

2. BERT in one page

- **BERT** (Bidirectional Encoder Representations from Transformers, Google 2018) is a stack of **Transformer encoder** layers. BERT-base: **12 layers, hidden size 768, 12 attention heads, about 110 M parameters**.
- **Bidirectional**: each token's vector is computed by attending to all other tokens on both sides, so the representation of 'cold' differs in 'I have a cold' vs 'the room is cold'. These are **contextual embeddings**.
- **Pre-training** (self-supervised, no labels): **Masked Language Modelling** - hide 15% of tokens and predict them; and **Next Sentence Prediction**. This teaches grammar and meaning from raw text.
- **Fine-tuning**: add a small task-specific layer on top and train the whole network on labelled data for a task (NER, classification, QA).
- Special tokens: **[CLS]** at the start, **[SEP]** at the end. Text is split into **WordPiece** sub-words ('migraine' -> 'mi', '##gra', '##ine').

3. What makes BioBERT different

	BERT-base (cased)	BioBERT (DMIS Lab, Korea Univ., 2019)
Starting point	random	initialised from BERT-base cased
Pre-training text	Wikipedia + BooksCorpus (3.3 B words)	+ PubMed abstracts (4.5 B words) + PMC full-text articles (13.5 B words)
Vocabulary	BERT WordPiece	same WordPiece vocabulary as BERT (kept for compatibility)
Architecture	12 layers, 768, 110 M params	identical
Strength	general English	biomedical terminology and phrasing

- BioBERT is a **domain-adapted language model**: same architecture, continued pre-training on biomedical text. It outperformed BERT on biomedical NER, relation extraction and question answering.
- **Base BioBERT has no NER ability by itself.** It only produces contextual vectors. To extract entities it must be fine-tuned with a token-classification head on a labelled NER dataset - which is what our two checkpoints are.

4. From BioBERT to an NER model: fine-tuning

```
BertForTokenClassification
  bert        = BioBERT encoder (12 Transformer layers)   -> (tokens, 768)
  dropout     = 0.1 during training (off at inference)
  classifier  = Linear(768 -> num_labels)                 -> (tokens, 3)  B-DIS, I-DIS, O

Training: cross-entropy loss between predicted and gold label for each token,
          back-propagated through the classifier AND all BioBERT layers.
```

Our checkpoints (by alvaroalon2, Apache-2.0):

Checkpoint	Fine-tuned on	Labels
alvaroalon2/biobert_diseases_ner	BC5CDR-disease (1,500 PubMed abstracts) + NCBI-disease (793 abstracts)	B-DISEASE, I-DISEASE, '0' (outside)
alvaroalon2/biobert_chemical_ner	BC5CDR-chemical + BC4CHEMD (10,000 PubMed abstracts)	B-CHEMICAL, I-CHEMICAL, O

5. BIO tagging and sub-word tokens

```
Text   : Have you had any nausea or vomiting ?
Tokens : [CLS] Have you had any nausea or vomit ##ing ? [SEP]
Labels :   -    O    O   O   O  B-DIS  O B-DIS I-DIS O   -
Entities: 'nausea' (DISEASE), 'vomiting' (DISEASE)
```

- **B-X** = beginning of an entity of type X, **I-X** = inside (continuation), **O** = outside. B is needed to separate two adjacent entities of the same type.
- **Sub-word rule**: a word takes the label of its **first sub-token** - the standard convention used when BERT NER models are trained. 'vomit ##ing' becomes the word 'vomiting'.
- **Character offsets** come from the fast tokenizer's offset_mapping, so each entity points exactly into the original sentence (we verified text[start:end] == entity text for every entity).
- **Confidence** = softmax probability of the predicted label, averaged over the entity's words. It is the model's real output, not an invented score - but neural networks are often over-confident (we see 1.00 a lot), so it is not a calibrated probability of being correct.
- Quirk: the disease checkpoint names its outside label '0' (zero) instead of 'O'; the code accepts both.

6. Our pipeline, step by step (clinical_extraction.py)

Step	Code	Output
Tokenizer	tokenizer(text, return_offsets_mapping=True)	WordPiece ids, char offsets, word ids
Encoder	model.bert(**enc).last_hidden_state	(1, tokens, 768) contextual vectors
Classifier	model.classifier(model.dropout(hidden))	(1, tokens, 3) logits
Softmax	logits.softmax(-1)	probability per label per token
Word labels	first sub-token of each word	label + probability per word
BIO decoding	B starts, I continues, O ends	entities with start/end/confidence

Step	Code	Output
Speaker context	attach role + diarization label + time	conversation[] in JSON

We call the encoder and classifier separately (instead of the one-line transformers pipeline) to make each stage visible; we verified the result is identical to calling the full model. Both models run on every utterance and their entities are merged and sorted by position. Input: outputs/<name>_roles.json or a text file of 'ROLE: text' lines. Output: outputs/<name>_clinical_entities.json. --raw prints every token's label and probability.

7. Model selection: what exists on Hugging Face

Checkpoint	Labels	Decision
siddharthtumre/biobert-finetuned-ner (most downloaded)	DNA, RNA, protein, cell_line, cell_type	rejected: molecular biology, not clinical
alvaroalon2/biobert_diseases_ner	DISEASE	**chosen**: named datasets, Apache-2.0
alvaroalon2/biobert_chemical_ner	CHEMICAL	**chosen**: drugs, Apache-2.0
ghadeermobasher/BC5CDR-Chemical-Disease...	Disease, Chemical	rejected: no model card or license
pmaitra/en_biobert_ner_symptom	symptom	rejected: spaCy wheel, unverifiable 99.95% F1
MACCROBAT models (many symptom/duration labels)	Sign_symptom, Duration, Severity ...	rejected: DistilBERT / DeBERTa, not BioBERT
dmis-lab/biobert-base-cased-v1.2	none (no NER head)	base model only - needs fine-tuning

8. Version 1 label mismatch: what we can and cannot extract

Project category	Supported?	How it appears
SYMPTOM	partly	under DISEASE - BC5CDR follows MeSH, which annotates signs and symptoms as diseases; symptom vs disease is not distinguished
DISEASE / CONDITION	yes	DISEASE
MEDICATION	mostly	CHEMICAL - includes drugs but also any chemical
DURATION	no	listed under unsupported_labels in the JSON
SEVERITY	no	listed under unsupported_labels in the JSON

> We deliberately **do not rename** DISEASE to SYMPTOM or CHEMICAL to MEDICATION: that would claim a capability the models do not have. Any later mapping must be an explicit, documented post-processing step.

9. Version 1 results (public BC5CDR checkpoints)

Utterance	Entities found
PATIENT: I have had fever for three days and a severe headache.	fever (DISEASE), headache (DISEASE)
DOCTOR: Do you have a cough?	cough (DISEASE)
PATIENT: I took paracetamol.	paracetamol (CHEMICAL)
PATIENT: My head has been hurting since yesterday, and I feel dizzy.	dizzy only - 'head ... hurting' missed
DOCTOR: Have you had any nausea or vomiting? PATIENT: Yes.	nausea, vomiting - but said by the DOCTOR
DOCTOR: It looks like a migraine. I will prescribe something for the pain.	migraine, pain (DISEASE)

- **Strong on medical nouns** (fever, headache, migraine, paracetamol): confidence about 1.00.
- **Weak on lay descriptions** ('my head has been hurting'): the models learned from scientific abstracts, not conversation - **domain shift**.
- **Speaker attribution is not symptom ownership**: the doctor said 'nausea' in a question; the patient confirmed with 'Yes.'. The SOAP stage must read question-answer pairs, not just entity speakers.
- 'Do you have a cough?' (doctor) and 'migraine' (doctor's assessment) show the same issue: an entity's speaker tells you its *role in the dialogue*, not whether the patient has it.

10. Evaluation of NER

- **Entity-level precision** = correct entities found / all entities found; **recall** = correct found / all true entities; **F1** = harmonic mean. Computed with the seqeval library.
- **Strict match**: same span and type. **Partial / lenient**: overlapping span and same type - useful when 'severe headache' vs 'headache' boundaries differ.
- Published BioBERT results (BC5CDR) are around 85-93 F1 on *abstracts*; on conversations expect lower. A proper evaluation needs a small hand-annotated set of our own consultation transcripts.

11. Limitations

- Domain shift: literature-trained models miss conversational phrasing.
- No negation / assertion: 'no fever' still yields fever. Needs a NegEx-style rule or an assertion model.
- No DURATION / SEVERITY / dosage / frequency labels.
- Symptom vs disease not separated; CHEMICAL is broader than medication.
- No entity linking: 'tummy ache' and 'abdominal pain' are not mapped to the same concept (UMLS / SNOMED CT). DMIS Lab's BioSyn models do this linking for BioBERT.
- Over-confident probabilities; errors from ASR (misheard drug names) pass straight through.
- 512-token limit per input (we process one utterance at a time, so not an issue here).

12. Version 2: fine-tuning BioBERT ourselves (Approach B)

12.1 Why, and the two options we compared

No public BioBERT checkpoint predicts duration or severity. Two ways to add them were considered:

Option	What	Pros	Cons
A. Rules	regex for 'for three days', 'since yesterday', 'mild/severe'; tagged source='rule'	fast, transparent	not BioBERT; brittle
B. Fine-tune BioBERT (chosen)	train dmis-lab/biobert-base-cased-v1.2 on MACCROBAT2018	genuine BioBERT predictions for all categories; great learning	training time; small dataset; case-report domain

12.2 Transfer learning in one picture

```
dmis-lab/biobert-base-cased-v1.2      (knows biomedical language, NO labels)
   + new classifier Linear(768 -> 15)   (random weights)
   + MACCROBAT labelled sentences
   -> train ALL weights for a few epochs with a small learning rate
   = models/biobert-maccrobat-ner       (BioBERT that predicts our 7 entity types)
```

- **Transfer learning**: reuse knowledge from pre-training (PubMed) and adapt it with little labelled data. Training BERT from scratch would need billions of words; fine-tuning needs only thousands of sentences.
- The classifier starts random, the encoder starts from BioBERT. Both are updated - this is **full fine-tuning** (alternative: freeze the encoder and train only the head, or LoRA adapters).

12.3 The dataset: MACCROBAT2018

- 200 English clinical **case reports** from PubMed Central, annotated by Caufield et al. (2019) with 41 entity types (Sign_symptom, Disease_disorder, Medication, Duration, Severity, Dosage, Lab_value, Age ...). License CC-BY-4.0. We used the Hugging Face copy ktgiahieu/maccrobat2018_2020 (JSONL of words + BIO tags).
- We kept 7 types: **Sign_symptom, Disease_disorder, Medication, Duration, Severity, Dosage, Frequency**. The other 34 types were set to O. Fewer, relevant labels = easier learning with little data. 15 labels in total: O + B-/I- for each type.
- Long reports were cut into sentences (at '.') and chunks of at most 120 words, closer to the length of a conversation turn and well under BERT's 512-token limit.

Type	Training examples in all 200 docs
Sign_symptom	4,505
Disease_disorder	1,759
Medication	1,446
Duration	409
Dosage	358
Frequency	68

Type	Training examples in all 200 docs
Severity	**40** - very few

12.4 Data problems we found (and fixed)

Problem	How we found it	Effect if not fixed	Fix
Every document appears **twice** (docs 0-199 = 200-399)	compared document texts	test documents also in training = **data leakage**, inflated scores	de-duplicate -> 200 docs
5,073 whitespace 'words' ('\n', thin spaces \u200a)	validation F1 stuck at 0.05 while train loss fell to 0.10; inspected alignment	tokenizer drops them -> gold labels shifted -> wrong scores	remove whitespace-only words; read gold labels at the same token positions as predictions
BioBERT 'cased' tokenizer lower-cases by default	tokenize('MRI') -> m ##ri	loses capitalisation (MRI, COVID, IgA)	do_lower_case=False

Lesson: when **training loss goes down but validation does not move at all**, suspect the data or the evaluation code before the model. A quick manual test on one sentence showed the model was actually fine.

12.5 Training setup (train_biobert_ner.py)

Choice	Value	Why
Split	160 / 20 / 20 documents (train / validation / test), seed 42	split by **document** so sentences of one report never appear in two splits
Labels for sub-words	first sub-token = word label; others = -100	-100 is ignored by PyTorch cross-entropy; same convention used at prediction time
Loss	token-level cross-entropy	standard for classification of each token
Optimiser	AdamW, lr 3e-5, weight decay 0.01	small lr so pre-trained knowledge isn't destroyed
Schedule	linear warm-up (10% of steps), then linear decay	warm-up avoids large, harmful updates while the new head is random
Gradient clipping	max norm 1.0	prevents occasional exploding updates
Batch / epochs	16 sentences / 5 epochs	about 2 min per epoch on Apple M4 (mps)
Model selection	keep the epoch with best **validation** F1	a form of early stopping
Final score	best checkpoint evaluated **once** on the 20 test documents	test data never used for decisions -> honest estimate

12.6 Learning curve

Epoch	Train loss	Validation F1	
1	0.531	0.333	
2	0.146	0.498	
3	0.098	**0.612**	best - saved
4	0.063	0.566	overfitting starts
5	0.043	0.572	

Train loss keeps falling but validation F1 drops after epoch 3: the model starts memorising training sentences (**overfitting**). Keeping the best validation epoch protects against it.

12.7 Test results (20 unseen case reports, entity-level, seqeval)

Label	Precision	Recall	F1	Test entities
Sign_symptom	0.561	0.653	0.603	219
Medication	0.551	0.808	0.656	73
Duration	0.586	0.739	0.654	23
Disease_disorder	0.360	0.429	0.391	63
Dosage	0.364	0.333	0.348	12
Severity	0.000	0.000	0.000	1
Frequency	-	-	-	0 (none in test)
Micro average	0.524	0.639	**0.576**	391

- Moderate scores are normal for MACCROBAT: it is small (160 training docs) and has fuzzy, long entity boundaries. Strict entity matching also counts 'severe abdominal pain' vs 'abdominal pain' as wrong.
- Severity and Frequency cannot be judged: 1 and 0 test examples. With only 40 Severity examples in all the data, the model essentially never predicts it.
- 20 test documents = a small, noisy estimate. Cross-validation would give a more stable number.

12.8 Version 1 vs Version 2 on our conversation

Utterance	Version 1 (BC5CDR)	Version 2 (our fine-tune)
I have had fever for three days and a severe headache.	fever, headache: DISEASE	fever, headache: Sign_symptom; **three days: Duration**; severe: missed
My head has been hurting since yesterday, and I feel dizzy.	dizzy: DISEASE	**hurting**, dizzy: Sign_symptom; since yesterday: missed
It looks like a migraine. I will prescribe ... for the pain.	migraine, pain: DISEASE	migraine: **Disease_disorder** (0.47); pain: missed
Just paracetamol, 500 milligrams, twice a day.	paracetamol: CHEMICAL (1.00)	milligrams, twice a day: Dosage; **paracetamol missed** here (found in 'I took paracetamol.')

Utterance	Version 1 (BC5CDR)	Version 2 (our fine-tune)
Are you taking any medication?	none	'medication': Medication (generic word)

- **Gains**: symptom vs disease separated, durations and dosages extracted, some lay wording caught, confidences no longer always 1.00.
- **Losses / gaps**: severity not learned, occasional missed drugs, generic words tagged.
- Both versions are kept: *--models maccrobat* (default) and *--models bc5cdr*.

12.9 How we could improve it

- More and closer data: annotate real doctor-patient transcripts (or use MTS-Dialog / ACI-Bench with our labels) - the biggest gain, since case reports are not conversations.
- Handle rare labels: oversampling sentences with Severity, or class-weighted loss.
- A **CRF layer** on top of BioBERT to enforce valid BIO sequences (no I- without B-).
- k-fold cross-validation and a hyperparameter search (lr, epochs, max length).
- Combine with Version 1: e.g. use the BC5CDR chemical model as a second opinion for drug names.

13. Alternatives to BioBERT

- **PubMedBERT / BiomedBERT** (Microsoft): pre-trained from scratch on PubMed with its own biomedical vocabulary; often slightly better than BioBERT.
- **ClinicalBERT / Bio_ClinicalBERT**: further trained on MIMIC-III clinical notes - closer to clinical language.
- **scispaCy** (en_ner_bc5cdr_md etc.): fast spaCy pipelines with UMLS linking.
- **MedCAT / cTAKES / MetaMap**: concept extraction with SNOMED/UMLS linking and negation.
- **LLMs**: zero-shot extraction of any category with a prompt, including duration and severity, but with privacy, cost and hallucination concerns; output must be validated against the text.

14. Interview questions and answers

Q1. Why did you fine-tune BioBERT yourself?

No public BioBERT NER checkpoint covered duration or severity, and BC5CDR merged symptoms with diseases. Fine-tuning base BioBERT on MACCROBAT2018 gave genuine predictions for Sign_symptom, Disease_disorder, Medication, Duration, Severity, Dosage and Frequency.

Q2. Walk me through your fine-tuning setup.

dmis-lab/biobert-base-cased-v1.2 plus a new Linear(768->15) head; 200 de-duplicated MACCROBAT case reports split 160/20/20 by document; sentences up to 120 words; first sub-token labelled, others -100; AdamW lr 3e-5, linear warm-up/decay, gradient clipping; 5 epochs on Apple M4; best validation F1 checkpoint; one final test.

Q3. What results did you get?

Test micro F1 0.58 (P 0.52, R 0.64). Medication 0.66, Duration 0.65, Sign_symptom 0.60, Disease 0.39, Dosage 0.35; Severity not learned (40 examples total, 1 in test).

Q4. Why split by document instead of by sentence?

Sentences from the same report share vocabulary and entities; splitting by sentence leaks information between train and test and inflates scores.

Q5. What is data leakage, and did you encounter it?

When information from the test set influences training. MACCROBAT's Hugging Face copy contained every document twice; a random split would have put copies of test documents in training. We de-duplicated first.

Q6. Your validation F1 was 0.05 while training loss fell. How did you debug it?

A manual prediction on one sentence was correct, so the model was fine. Inspection showed whitespace tokens ('\n', thin spaces) that the tokenizer drops, shifting gold labels against predictions. We removed them and read gold labels at the same token positions; F1 jumped to 0.61.

Q7. What is the -100 label?

PyTorch's cross-entropy ignore_index. We put it on [CLS], [SEP], padding and non-first sub-tokens so only one prediction per word contributes to the loss.

Q8. Why a learning-rate warm-up?

The new head is random at the start; large early updates would push random gradients into the pre-trained encoder. Warm-up starts small and increases, then decays linearly.

Q9. How did you detect and handle overfitting?

Train loss kept dropping while validation F1 peaked at epoch 3 then fell. We save the best validation checkpoint (early stopping) and only evaluate that on the test set.

Q10. Why might cased vs uncased matter?

BioBERT v1.2 is cased but its tokenizer config lower-cases by default. Capitalisation distinguishes MRI, COVID, IgA from ordinary words, so we set do_lower_case=False for training and inference.

Q11. How would you improve the fine-tuned model?

Annotated conversational data, oversampling/class weights for rare labels, a CRF layer, cross-validation and hyperparameter search, and ensembling with specialised drug models.

Q12. Isn't mapping Sign_symptom to SYMPTOM just relabelling?

It's a fixed 1-to-1 renaming of labels the model was trained on, stored separately as 'category' while 'label' keeps the model's own name - unlike renaming BC5CDR's DISEASE to SYMPTOM, which would be a guess.

Q13. What is BioBERT?

BERT-base initialised from Google's weights and further pre-trained on PubMed abstracts and PMC articles. Same architecture (12 layers, 768 hidden, 110M parameters) and vocabulary; better biomedical language understanding.

Q14. Can base BioBERT extract symptoms directly?

No. It outputs contextual vectors only. You need a token-classification head fine-tuned on a labelled NER dataset. We used fine-tuned BertForTokenClassification checkpoints.

Q15. Walk me through your extraction pipeline.

Tokenizer -> WordPiece ids and offsets; BioBERT encoder -> 768-d contextual vector per token; linear classifier -> logits over BIO labels; softmax -> probabilities; first-sub-token rule -> word labels; BIO decoding -> entities with character offsets and confidence; attach speaker role.

Q16. What is BIO tagging and why the B tag?

B = beginning, I = inside, O = outside. B separates adjacent entities of the same type, which plain IO tagging cannot.

Q17. How do you handle WordPiece sub-words?

Each word takes the label of its first sub-token (the training convention), and entity spans use the tokenizer's offsets so 'vomit ##ing' maps back to 'vomiting'.

Q18. Which models did you use and why?

alvaroalon2/biobert_diseases_ner and biobert_chemical_ner: genuine fine-tuned BioBERT, documented datasets (BC5CDR, NCBI-disease, BC4CHEMD), Apache-2.0. The most-downloaded BioBERT NER model was for proteins/DNA.

Q19. Why do symptoms come out as DISEASE?

BC5CDR follows MeSH, where signs and symptoms are annotated as disease mentions. So the model finds them but can't distinguish symptom from disease.

Q20. Why didn't you relabel DISEASE as SYMPTOM?

It would misrepresent the model's capability. Mappings must be explicit post-processing steps that can be evaluated, otherwise downstream clinical notes contain unverified claims.

Q21. How would you extract duration and severity?

Short term: transparent rules tagged as rule-based. Properly: fine-tune BioBERT on MACCROBAT2018 which has Duration, Severity, Sign_symptom, Medication labels, and evaluate with seqeval.

Q22. How is the confidence computed and can you trust it?

Average softmax probability of the predicted label across the entity's words. Real model output, but neural networks are over-confident; calibration (e.g. temperature scaling) would be needed to treat it as a probability of correctness.

Q23. Why did the model miss 'my head has been hurting'?

Domain shift: trained on scientific abstracts with terms like 'headache', not lay conversational descriptions. Fine-tuning on conversational clinical data would help.

Q24. Why keep the speaker with each entity?

SOAP needs provenance: patient-reported (Subjective) vs clinician-stated (Assessment/Plan). But speaker isn't ownership - a doctor's question 'any nausea?' answered 'yes' means the patient has nausea.

Q25. What is negation detection and why does it matter?

Deciding whether a mentioned finding is present or absent ('denies chest pain'). Without it, 'no fever' would be recorded as fever. NegEx rules or assertion classifiers are common solutions.

Q26. What is entity linking?

Mapping a mention to a concept ID in an ontology (UMLS, SNOMED CT, RxNorm) so synonyms unify and the data is interoperable. BioSyn and SapBERT are BioBERT-family linkers.

Q27. How would you evaluate this stage?

Annotate a set of consultation transcripts, compute entity-level precision, recall and F1 (strict and partial) per label with seqeval, and analyse errors by type (lay terms, negation, ASR errors).

Q28. BioBERT vs PubMedBERT vs ClinicalBERT?

BioBERT: BERT continued on PubMed/PMC with BERT's vocab. PubMedBERT: from scratch on PubMed with a domain vocab. ClinicalBERT: further trained on clinical notes (MIMIC). Choice depends on whether text is literature-like or clinical-note-like.

Q29. Why run two models instead of one?

No well-documented, licensed BioBERT checkpoint covered both diseases and chemicals; two specialised models are documented and Apache-2.0. Cost: two forward passes and ~860 MB of weights.